Group 1: Morning Book Study (Alternate Focus Areas)

You will alternate between the following books each morning, focusing on key topics and taking notes:

1. **Valvano's Book**:
   - Read the first 40 pages (introductory concepts on computers and electronics).
   - Ongoing study of embedded systems, focusing on how it ties into your practical projects.
2. **Yiu's "Embedded Systems"**:
   - Focus on chapters related to ADC and microcontroller-specific concepts.
   - Supplement Valvano's book for hardware configuration and register-level understanding.
3. **"The Art of Electronics"**:
   - Supplement Valvano with foundational electronics concepts, especially on signals, sinusoidal functions, and ADC.
   - Focus on introductory chapters that explain electronics fundamentals, and reference page 14 on sinusoidal signals.
4. **Brown's Book**:
   - Use as a reference for minimal projects and automation, particularly ADC and hardware setup.
   - Supplement Yiu and Valvano for specific topics like ADC and signal processing.
5. **CLRS and Sedgewick**:
   - Continue with your focus on sliding window algorithms, KMP, Rabin-Karp, tries, and B-trees.
   - De-prioritize the rest of the data structures since you're already familiar with them.

Group 2: Follow-Up on FFT Applied to ADC

1. **Review Polynomials and Coefficients in Relation to FFT**:
   - **Reference Books**: CLRS and Sedgewick. Review sections on polynomials and their role in algorithms, focusing on their application in FFT.

- **Action**: Understand how polynomials form the basis for the FFT algorithm (e.g., how FFT decomposes a signal into sinusoidal components).
- **Goal**: Grasp the mathematical foundation (polynomials, coefficients) that underpins the FFT.

2. **Basic Electronics and ADC Fundamentals**:
   - **Reference Books**: The Art of Electronics, Valvano's Book, Yiu's "Embedded Systems", and Brown's Book.
   - **Action**: Map out how the ADC works, particularly in the STM32 context. Extract formulas from the books and locate corresponding sections in the STM32 datasheets/manuals.
   - **Goal**: Develop a clear understanding of ADC operations and their connection to signal processing with FFT.

3. **Sinusoidal Signals and Magnitude**:
   - **Reference**: The Art of Electronics, page 14.
   - **Action**: Learn the meaning of "magnitude" in the context of FFT and how it represents frequency components.
   - **Goal**: Understand the breakdown of sinusoidal signals via FFT and the interpretation of magnitude.

4. **FFT Algorithm Based on Sedgewick**:
   - **Reference**: Sedgewick's Algorithms.
   - **Action**: Continue using Sedgewick's FFT implementation in your project. Verify the algorithm with real-time ADC data.
   - **Goal**: Ensure the FFT algorithm works effectively in your project.

5. **Research the CMSIS FFT Library (for reference only)**:
   - **Action**: Explore CMSIS FFT to familiarize yourself with the professional tools available for future reference.
   - **Goal**: Gain awareness of how CMSIS libraries could be leveraged in the future.

6. **Formulas and Practical Applications**:
   - **Action**: Gather key formulas related to ADC (e.g., converting ADC values to voltages, scaling signals for FFT input) and confirm their presence in the STM32 manual.
   - **Goal**: Have practical formulas ready for your project.

Group 3: Algorithms and Data Structures

1.1 Algorithms

- **Fast Fourier Transform (FFT)**: Study the algorithm for frequency domain analysis, focusing on efficient computation in real-time systems.
- **Sliding Window Algorithm**: Used for maintaining a subset of items (e.g., moving average or window-based data aggregation) in real-time data streams.
- **Pattern Matching**:
  - KMP Algorithm: Knuth-Morris-Pratt for efficient string matching.
  - Rabin-Karp Algorithm: Used for detecting patterns in streams of data.

1.2 Data Structures

- **Trie**: Efficient for storing and querying dynamic sequences or strings.
- **B-tree**: Balanced tree structure used for large datasets.

2. Integrated Embedded Systems Tasks (Hardware + Algorithms)

2.1 Priority 1: Foundation-Level Tasks

- **Temperature and Voltage Monitoring**
  - Data Source: Internal temperature sensor, voltage reference (VREFINT)
  - Data Structures: Ring Buffer, Priority Queue
  - Algorithms: Moving Average, Exponential Smoothing, Threshold Detection
- **Watchdog Timers**
  - Data Source: Independent and Window Watchdog Timers (IWDG, WWDG)
  - Data Structures: Circular Queue
  - Algorithms: Timeout Management, Monitoring and Alerting
- **Real-Time Clock (RTC) & Timestamps**
  - Data Source: RTC, SysTick timer
  - Data Structures: Binary Search Tree, B-tree
  - Algorithms: Scheduling Algorithms (Round-robin, priority-based), Binary Search

2.2 Priority 2: Intermediate-Level Tasks

- **Interrupt-Driven Task Scheduling**
  - Data Source: Timers, interrupts
  - Data Structures: Priority Queue, Doubly Linked List
  - Algorithms: Round-Robin Scheduling, Earliest Deadline First (EDF)
- **Real-Time Signal Processing with ADC**
  - Data Source: Analog-to-Digital Converter (ADC)
  - Data Structures: Circular Queue, FFT Buffer
  - Algorithms: FFT, Sliding Window, Finite State Machine (FSM) for pattern recognition
- **Fault Monitoring and Recovery**
  - Data Source: Non-Maskable Interrupt (NMI), system faults
  - Data Structures: HashMap, Circular Queue
  - Algorithms: Error Detection (Hamming Code), CRC

2.3 Priority 3: Advanced-Level Tasks

- **DMA for Data Streaming**
  - Data Source: DMA controller, ADC
  - Data Structures: Ring Buffer, Trie
  - Algorithms: Pattern Matching (KMP, Rabin-Karp), Circular Buffer Management
- **Memory Management and Flash Programming**
  - Data Source: Flash memory, EEPROM
  - Data Structures: HashMap, Linked List
  - Algorithms: Wear-Leveling, CRC for Data Integrity
- **Power Consumption Monitoring**
  - Data Source: Internal power sensors
  - Data Structures: Priority Queue, HashMap
  - Algorithms: Power Optimization (Dynamic Voltage/Frequency Scaling), Energy Profiling